TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 3

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a basic calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

Exam paper

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: Past Paper 3 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [MI/AI]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 34 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Section A	15		
Section B	45		
Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 0.238 + 0.05

Calculate: 0.72 × 0.4 = 0.288

Answer	0.288 ✓ [1 mark]

[1]

2. Work out 0.8 × 0.222

Calculate: 1.48 × 0.12 = 0.1776

Answer	0.1776 ✓ [1 mark]

[1]

Past paper

Questions continue on the following page

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3. A company has 200 employees and 2 offices.

Employees are split across both offices as shown in the table below:

	Office A	Office B
Part-time employees	64	48
Full-time employees	16	72

One person is chosen at random.

What is the probability that the person chosen works full time in office B?

Give your answer as a percentage.

Step 1: Probability = 72 / 200

Percentage: (72 ÷ 200) × 100 = 36%

Answer 36% ✓ [2 marks] %

[2]

4. Round 39.4528 to the nearest whole number.

Use your rounded number to **estimate** the answer to 39.4528 × 2

Show your working.

Round 39.453 to 2 significant figures = 39

Double the rounded number: 39 × 2 = 78

Answer 78 ✓ [2 marks]

[2]

Past paper

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5. Idris sees this mobile phone in a sale:

Mobile phone
Was £95.00
Now 12% off

How much money will Idris save buying the phone in the sale?

Step 1: Discount = 12% of £95

Calculation: 95 × 0.12 = £11.40

Answer £ **£11.40** ✓ *[2 marks]*

[2]

6. Work out $10^2 – 25 \times 4$

BIDMAS: Powers and Multiplication before Subtraction

$10^2 = 100$ and $25 \times 4 = 100$ --> $100 - 100 = 0$

Answer **0** ✓ *[2 marks]*

[2]

Past paper

Questions continue on the following page

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7. Work out:

$$5\frac{1}{3}+4\frac{1}{5}$$

Give your answer as a mixed number.

Common denominator of 3 and 5 is 15:

5 1/3 = 5 5/15 and 4 1/5 = 4 3/15

Total = (5 + 4) + (5/15 + 3/15) = 9 8/15

Answer	9 8/15 ✓ [2 marks]

[2]

8. Alex has just been on a cruise:

- the cruise ship travelled 4320 miles
- it took 10 days of non-stop sailing.

Alex thinks the average speed of the cruise ship was 20 miles per hour.

Is Alex correct?

Show how you decide.

Total hours sailing = 10 days × 24 hours/day = 240 hours

Average speed = 4320 miles ÷ 240 hours = 18 mph

Comparison: 18 mph < 20 mph. Therefore No.

Answer	No (18 mph ✓ [3 marks]

[3]

End of Section A.

Past paper

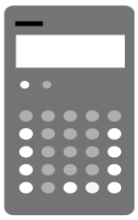
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Section B begins on Page 6.

Past paper

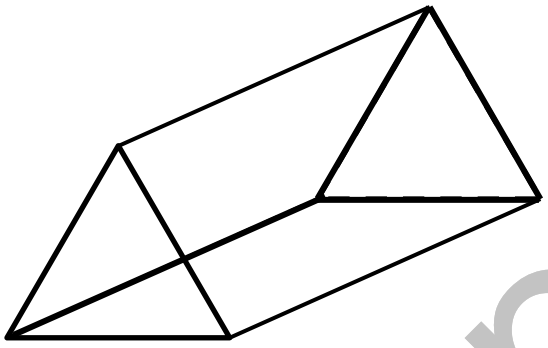
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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. What is this shape?



Answer	Triangular prism ✓ [1 mark]

[1]

2. Put these numbers in order starting with the lowest:

1 025 600 –102 560 –85 956 1 102 560

Answer	-102 560 , -85 956 , 1 025 600 , 1 102 560 ✓ [1 mark]		
	Lowest →		**Highest**

[1]

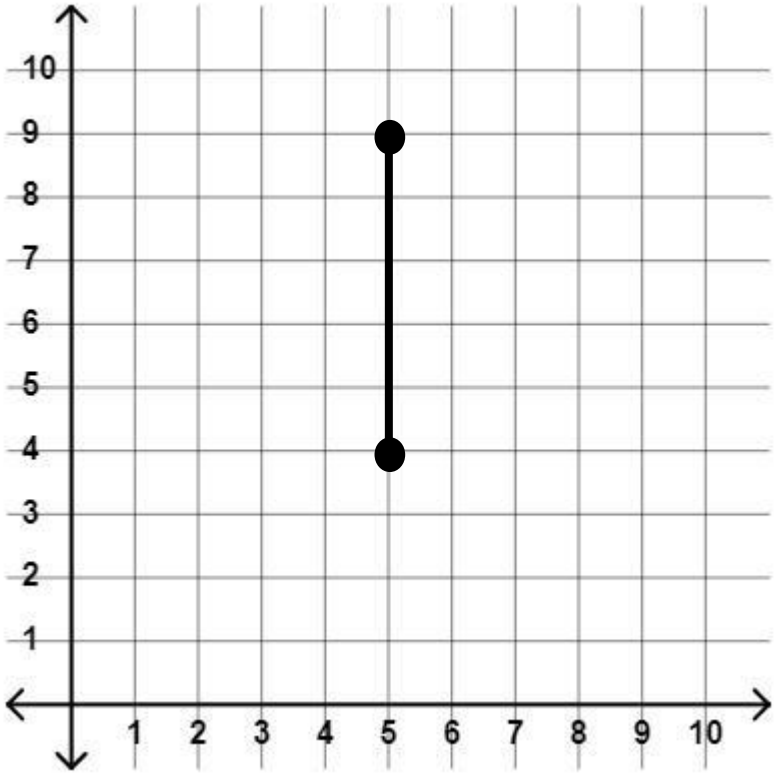
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3. Ezra is going to put a shed in a garden.

The shed is in the shape of a square.

The coordinate grid shows where two corners and one edge of the square will go.



What are the possible coordinates for the other **two** corners of the square?

Shed width = 2 m (4 squares), length = 3 m (6 squares)

Opposite vertices from (6, 4) and (6, 9) --> (10, 4) and (10, 9)

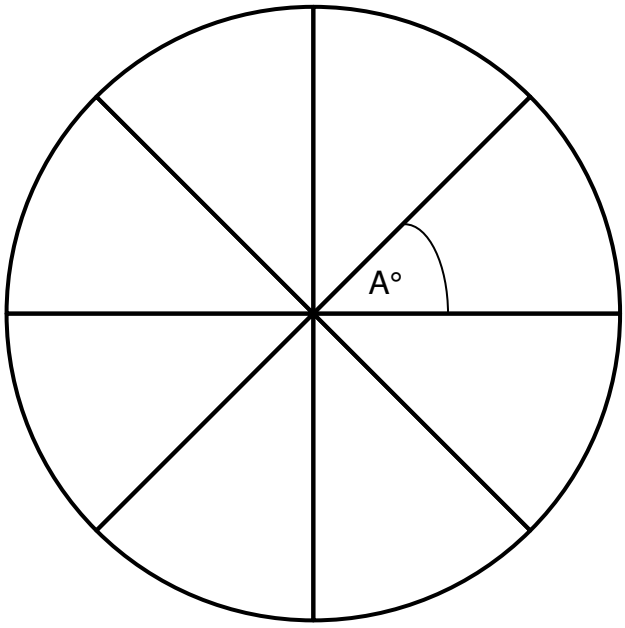
Answer (10, 4) and (10, 9) [2 marks]

[2]

Questions continue on the following page

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4. A circle is divided into 8 equal sections as show below:



Calculate the value of A.

Full circle = 360°. Divided into 8 equal sectors:

Angle of each sector = 360° ÷ 8 = 45°

Answer	45° ✓ [2 marks] °

[2]

Past pap

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5. Write this fraction:

$$\frac{29}{40}$$

(a) as a decimal

(b) as a percentage.

Fraction 29/40: 29 ÷ 40 = 0.725

Percentage: 0.725 × 100 = 72.5%

Answer	**(a) Decimal:** 0.725	**(b) Percentage:** 72.5% ✓ [2 marks]

[2]

Past paper

Questions continue on the following page

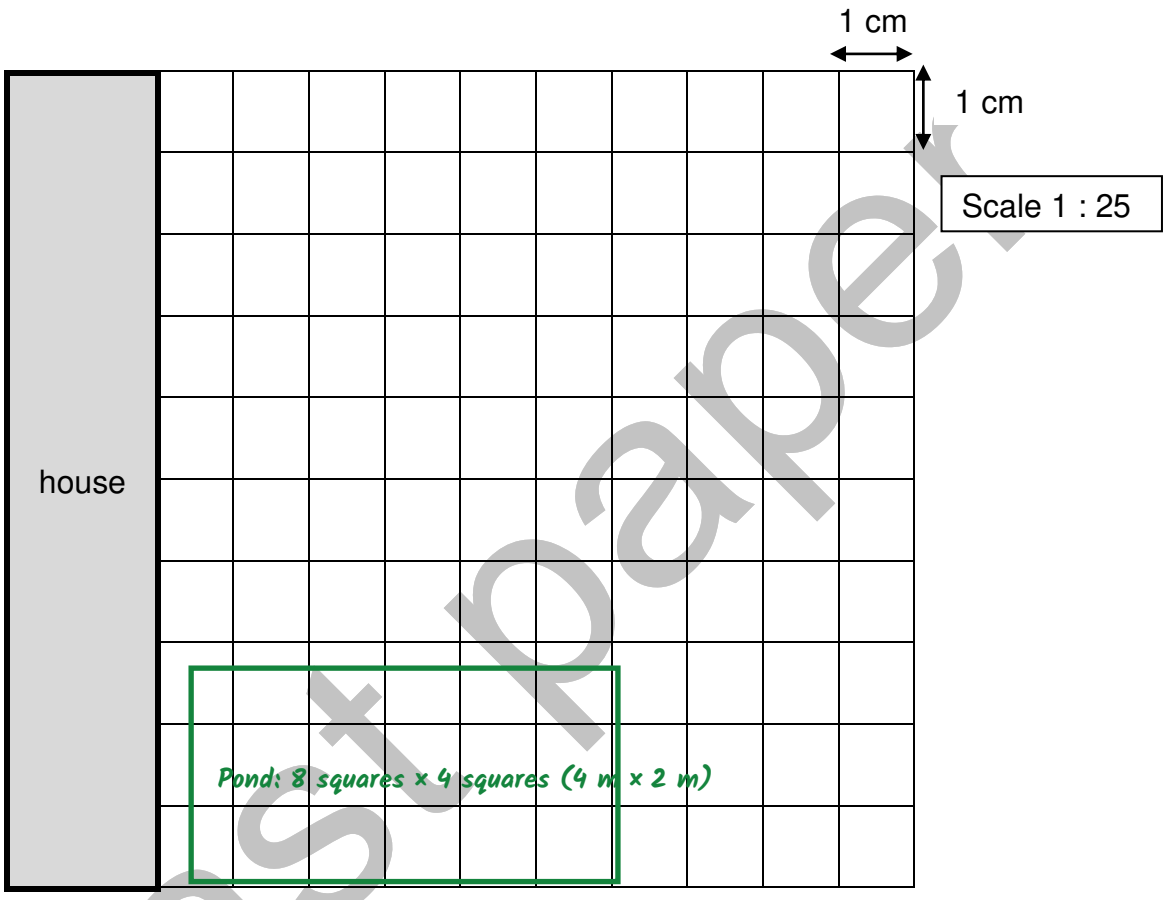
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6. Charlie wants to build a pond in a garden.

The pond needs to be a rectangle with:

- a length of 2 metres and a width of 1 metre
- **at least** 1 metre away from the house.

Draw a possible position for the pond on the scale diagram below:



[2]

Past paper

Scale: 1 square = 0.5 m --> 4 m = 8 squares, 2 m = 4 squares

Positioned at least 2 m (4 squares) away from the house

[2 marks]

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7. Last year 825 500 people visited a museum.

The manager is set a target to increase the number of visitors by 22%.

The manager thinks this will be more than 1 005 000 visitors.

Is the manager correct?

Show how you decide.

	Step 1: Increase by 22% = × 1.22 825 500 × 1.22 = 1 007 110 Comparison: 1 007 110 > 1 005 000. Therefore Yes.
Answer	Yes (1 007 110 > 1 005 000) [3 marks]

[3]

Past paper

Questions continue on the following page

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8. Leslie needs to put a fence around a pond.

The pond is in the shape of a circle with a diameter of 4.26 meters.

Leslie buys 13.5 metres of fencing.

Has Leslie bought enough fencing?

Show how you decide.

Use π = 3.14

Formula: Circumference = π × d

Circumference = 3.14 × 4.26 m = 13.3764 m

Comparison: 13.3764 m < 13.5 m. Therefore Yes, Leslie has enough.

Answer Yes (13.38 m < ✓ 13.5 m) [3 marks]

[3]

Past paper

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9. Taylor books a holiday.

The next day, their friend books the same holiday for £690 **after** receiving a 4% discount.

How much more did Taylor pay for the holiday?

Step 1: £690 represents (100% - 4%) = 96% of original price

Original price = £690 ÷ 0.96 = £718.75

Money saved = £718.75 - £690 = £28.75

Answer £ £28.75 [3 marks]

[3]

Past paper

Questions continue on the following page

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10. Chen sells cakes.

The table below shows the ingredients in one cake:

Ingredient	Grams
Flour	275
Sugar	200
Baking powder	5

A customer wants to know what fraction of the cake is sugar.

Chen tells the customer $\frac{7}{12}$

Is this correct?

Show how you decide.

Total ingredients = 275 g + 200 g + 5 g = 480 g

Fraction of butter = 200 ÷ 480 = 5/12

Comparison: 5/12 ≠ 7/12 (5/12 is less than 7/12). Therefore No.

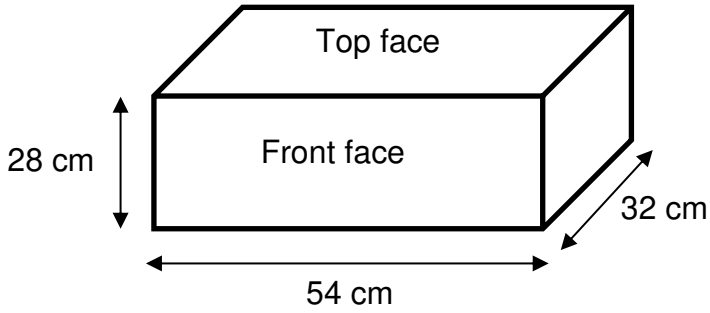
Answer	No (fraction is ...) [3 marks]

[3]

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11. Lee wants to paint the front face and top face of this cuboid as part of an art project:



Not drawn to scale

Lee will paint yellow, green and white stripes in the ratio 5 : 3 : 2

How much of the cuboid will be painted yellow?

Give your answer in square centimetres (cm^2).

Base = 54 × 28 = 1512 cm^2 / 2 sides = 2 × (32 × 28) = 1792 cm^2

Total Surface Area = 3240 cm^2

Half-area = 3240 ÷ 10 × 5 = 1620 cm^2

Answer	1620 cm^2 ✓ [4 marks] cm^2

[4]

Past paper

Questions continue on the following page

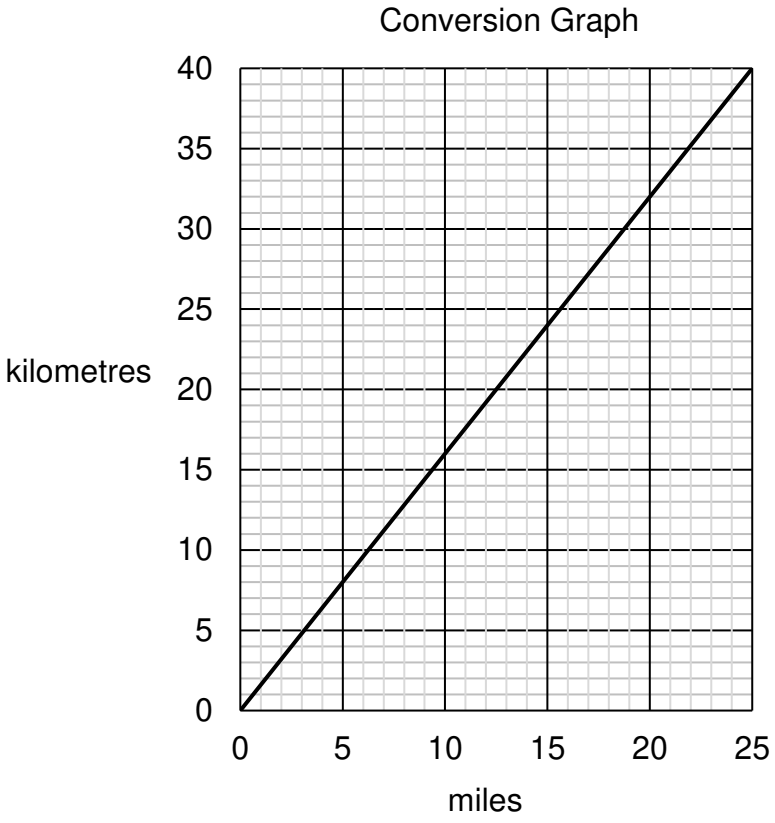
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12. Harper is planning a road trip with this information:

- the total distance is 1600 **kilometres**
- Harper only wants to drive a maximum of 225 **miles** each day.

How many whole days will Harper need to complete the 1600 km?

Use this graph to convert between kilometres and miles.



Show your working.

Conversion from graph: 8 km = 5 miles (or 40 km = 25 miles)

1600 km = (1600 ÷ 8) × 5 = 1000 miles

Days = 1000 ÷ 225 = 4.444... days --> Rounds up to 5 days

Answer	5 days ✓ [4 marks] **days**

[4]

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13. Over a period of six weeks, Jamie sold chocolate cupcakes and lemon cupcakes.

The number of chocolate cupcakes Jamie sold each week is shown below:

Number of chocolate cupcakes sold					
Week 1	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**
91	85	64	68	100	72

The number of lemon cupcakes Jamie sold is summarised below:

Lemon cupcakes sold	
Mean number sold per week	84
Range	20

Jamie claims:

a) "On average, I have sold more chocolate cupcakes per week"

b) "The chocolate cupcake sales are more consistent".

Are both of Jamie's claims correct?

Give reasons for your answers.

Show working here:
Reason - claim a):
Reason - claim b):

[4]

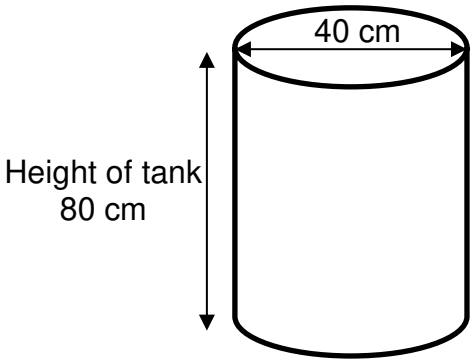
Questions continue on the following page

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14. Ali is setting up a saltwater fish tank.

The fish tank is in the shape of a cylinder.



Not drawn to scale

Ali uses this formula to work out how many grams of salt to add to the tank:

$$\text{Amount of salt in grams} = \left(\frac{\text{volume of tank}}{1000}\right) \text{multiplied by } 35$$

Ali thinks the tank will need more than 3.75 kilograms of salt.

Is Ali correct?

Show how you decide.

Use π = 3.14

Mean for Lemon: (91+85+68+104+84+48) ÷ 6 = 480 ÷ 6 = 80

Range for Lemon: 104 - 48 = 56 (Orange range is 70 - 45 = 25)

Comment: No, Lemon had a higher mean (80 > 56) and higher variation.

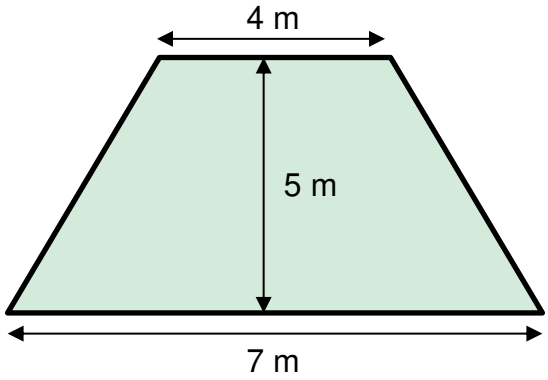
Answer No (Mean = 80, R ✓ 56) [4 marks]

[5]

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15. Riley wants to estimate how much it will cost to lay a patio in the garden below:



Not drawn to scale

Riley finds these prices online:

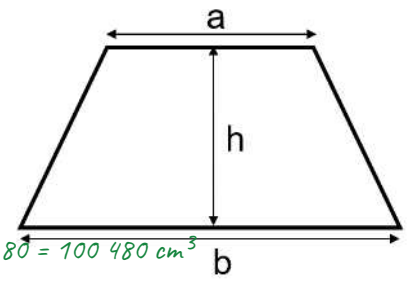
Price per square metre (m^2)					
£85	£100	£75	£130	£150	£135

Riley uses the median price to estimate how much it will cost to lay the patio.

How much does Riley estimate the patio will cost?

Use:

$$\text{Area} = \frac{(a+b)h}{2}$$



Cylinder Volume = $\pi \times r^2 \times h = 3.14 \times 20^2 \times 80 = 100\,480\ cm^3$

Mass = (100 480 ÷ 1000) x 35 g = 3516.8 g = 3.5168 kg

Comparison: 3.5168 kg < 3.75 kg. Therefore No.

Show your working.

Answer box is on the next page

Past Paper

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Q14 Conclusion: 3.517 kg < 3.75 kg --> No

No (3.52 kg < 3.75 [5 marks]

Trapezium Area = ((4 + 7) ÷ 2) × 5 = 5.5 × 5 = 27.5 m^2

Ordered costs: 75, 85, 100, 130, 135, 150 --> Median = (100 + 130) ÷ 2 = £115

Total Cost = 27.5 m^2 × £115 = £3162.50

Answer	£ £3162.50 [6 marks]

[6]

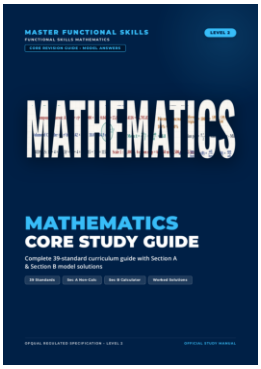
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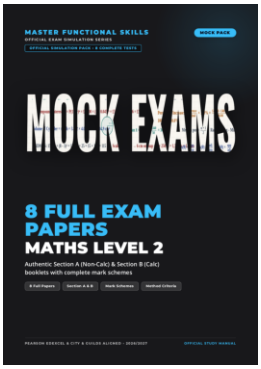


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